Income Targeting and Segmentation (Report)

1. Summary of the Project:

This report presents a complete data science solution for identifying and targeting individuals likely to earn above or below $50,000 annually, giving your marketing team two well-defined audiences to work with. Using weighted census survey data from 1994 and 1995, I trained and evaluated three classification models, built a customer segmentation framework, and designed a profit-optimised decision threshold that tells you exactly who is worth contacting and at what scale.

The recommended model is XGBoost, operating at a classification threshold of 0.177 rather than the standard default of 0.5. At this threshold, the model flags approximately 9.7% of the population for outreach, reaching a meaningful share of high-income individuals while keeping marketing costs manageable. On the held-out test set, XGBoost achieves a ROC-AUC of 0.948 and a PR-AUC of 0.649, a substantial improvement over both the logistic regression baseline and random forest.

Race, sex, and hispanic origin are excluded from the feature set entirely, and it is worth explaining why. These variables do correlate with income in the raw data, but that correlation reflects decades of structural inequality baked into the labour market of the mid-1990s rather than any genuine difference in individual earning capacity. A model trained on them would quietly encode that history into future campaigns. The model described in this report predicts income from employment behaviour, education, investment activity, and work patterns, factors that reflect what people do, not who they are.

2. Data Understanding and Exploration

a. Dataset Overview

The dataset comes from the U.S. Census Bureau's Current Population Surveys conducted in 1994 and 1995. Each record contains 40 demographic and employment-related variables for an individual, along with a sample weight and a binary income label: whether that person earns above or below $50,000 per year. The full dataset contains 199,523 records after loading.

One variable that required immediate attention is the sample weight. Because this data comes from a stratified survey design, each record does not represent one person — it represents a population count. A record with a weight of 1,000 effectively stands in for 1,000 people in the general population with similar characteristics. Throughout this analysis, all reported metrics and charts use population-weighted counts rather than raw sample counts. For a retail client making real-world marketing decisions, the weighted figures are what actually matter.

b. Class Imbalance

The most structurally important feature of this dataset is its class imbalance. In the raw sample, 93.8% of records are labelled as earning below $50K, with only 6.2% above it. In the population-weighted counts, this ratio is nearly identical: 93.6% below and 6.4% above.

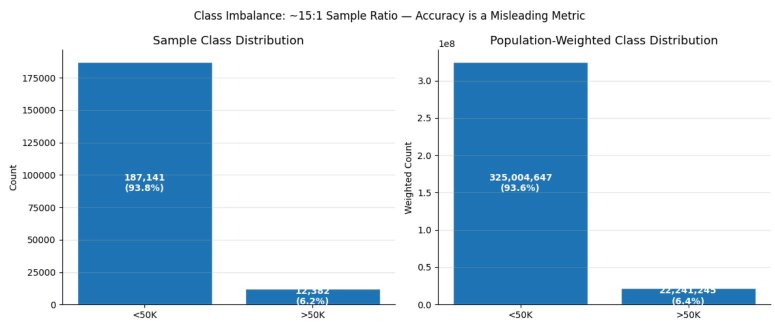

Figure 1. Class distribution by raw sample count (left) and population-weighted count (right)

This imbalance has direct consequences for model training and evaluation. A classifier that simply predicts 'under $50K' for every record would achieve 93.8% accuracy while being completely useless for marketing purposes. For this reason, accuracy is never used as an evaluation metric in this project. Instead, the primary metric is Precision-Recall AUC (PR-AUC), which is specifically designed for imbalanced settings and penalises models that ignore the minority class.

c. Fairness: Race and Sex

Before deciding how to handle race, sex, and hispanic origin, I first examined whether these variables actually carry income signals in the data. They do, and that is precisely the problem. The income gaps visible in the raw data between demographic groups are not evidence of differential earning capacity. They reflect the labour market conditions of 1994–95, when structural inequality kept wages lower for women and many minority groups regardless of their qualifications or work patterns.

If these variables were fed into the model, it would learn these historical correlations and embed them into future predictions. In practice, your marketing campaigns would end up systematically under-targeting women and certain racial groups, not because they are less likely to be high earners, but because the training data comes from a time when they were paid less. That is not a model worth deploying.

Decision: race, sex, and hispanic origin are excluded from the feature set. This is enforced in the project configuration file and takes effect before any feature engineering or model training. The model predicts income from employment behaviour, education, investment activity, and work intensity, not demographic identity.

3. Data Preprocessing & Feature Engineering

a. Cleaning Steps

Several columns were dropped before modelling: the year variable (constant across the dataset), the detailed industry and occupation recode columns (redundant with the major code versions), the sample weight (extracted separately and never used as a feature), and the three protected attribute columns described in Section 2.b.

Missing values in the migration-related columns were handled carefully. Rather than importing them immediately, I first created a binary flag, migration_data_absent — that captures whether all migration columns are missing for a given record. This is meaningful because missingness here likely indicates the person did not move, which is itself a signal. After flagging, missing values in migration columns and all other string columns were filled with 'Unknown', preserving them as a distinct category rather than discarding the records.

Numeric columns with skewed distributions, capital gains, capital losses, dividends from stocks, and wage per hour, were log-transformed using log1p after clipping to zero. This reduces the influence of extreme values without discarding them.

b. One-Hot Encoding

All categorical variables were one-hot encoded. The encoding vocabulary was fitted on the training set only, and validation and test sets were reindexed to match exactly, ensuring no information from those sets influenced the encoding. drop_first was set to False for tree-based models (Random Forest and XGBoost), where keeping all dummy levels improves feature importance interpretability. For logistic regression, regularisation handles the resulting collinearity without needing to drop a level.

c. Business-Informed Features

Beyond standard preprocessing, I engineered three additional features based on real-world income patterns:

• prime_working_age: a binary flag for individuals aged 25–55, capturing peak earning years. Wages are typically highest in this range, past entry-level and before retirement transitions begin.

• is_retirement_age: a flag for individuals aged 62 and over, the Social Security eligibility threshold in the US. At this age, wage income often shifts to fixed or investment income, which changes the prediction signal.

• capital_activity_flag: a binary flag indicating whether a person has any capital gains or stock dividend income. This collapses two sparse financial columns into one interpretable signal that proved strongly predictive of higher income.

Each feature was validated by checking population-weighted income rates by group before committing it to the pipeline. All three showed meaningful separation between income classes and were carried forward into model training.

4. Model Architecture & Training

a. Model Selection Rationale

Three models were trained: **Logistic Regression, Random Forest, and XGBoost**. This was a deliberate choice rather than just trying everything available. Logistic Regression serves as the interpretable baseline — if it performs well enough, there is no justification for the added complexity of an ensemble model. Random Forest and XGBoost are both strong ensemble methods, but they have different characteristics: Random Forest tends to be more stable with default hyperparameters while XGBoost typically achieves higher accuracy through its boosting approach but is more sensitive to tuning.

b. Data Splits and Imbalance Handling

The data was split into 70% training, 15% validation, and 15% test sets, with stratification on the income label to preserve the class ratio across all splits. The validation set was used for threshold selection and profit simulation, it never touched the model during training. The test set was held out entirely until final evaluation.

c. Training Results

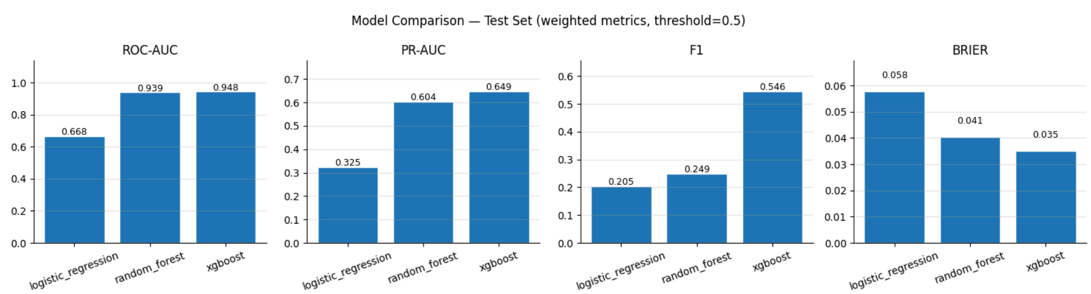

Figure 2. Model comparison on the test set across four metrics. PR-AUC is the primary metric. XGBoost leads on all four, with Random Forest close behind on ROC-AUC.

XGBoost achieves a test PR-AUC of 0.649 against a weighted baseline of 0.064, roughly 10x better than random. Its ROC-AUC of 0.948 confirms strong overall discrimination. Random Forest is competitive on ROC-AUC (0.939) but falls further behind on PR-AUC (0.604), suggesting it is less precise when identifying the minority class. Logistic Regression, while interpretable, shows considerably weaker performance across all metrics, making it unsuitable as the production model for this use case.

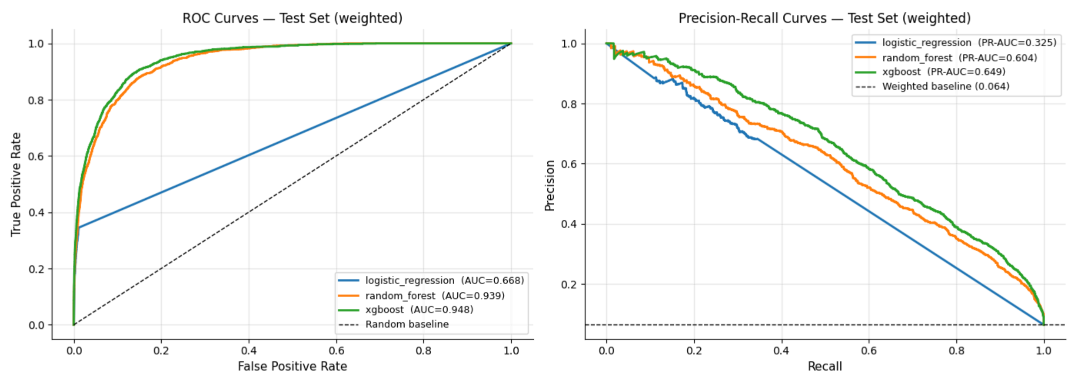

Figure 3. ROC curves (left) and Precision-Recall curves (right) for all three models on the test set. The gap between XGBoost and Logistic Regression is most visible in the PR curve, particularly at lower recall values where marketing precision matters most.

5. Evaluation & Threshold Selection

a. Why the Default Threshold is Wrong

Classification models output a probability score between 0 and 1. By default, a threshold of 0.5 is used to convert this into a binary prediction. This default assumes classes are roughly balanced, which they are not here. With only 6.4% of the population earning above $50K, a threshold of 0.5 causes the model to be far too conservative, missing many genuine high-income individuals who receive scores below 0.5 simply because the model is calibrated to a sparse class.

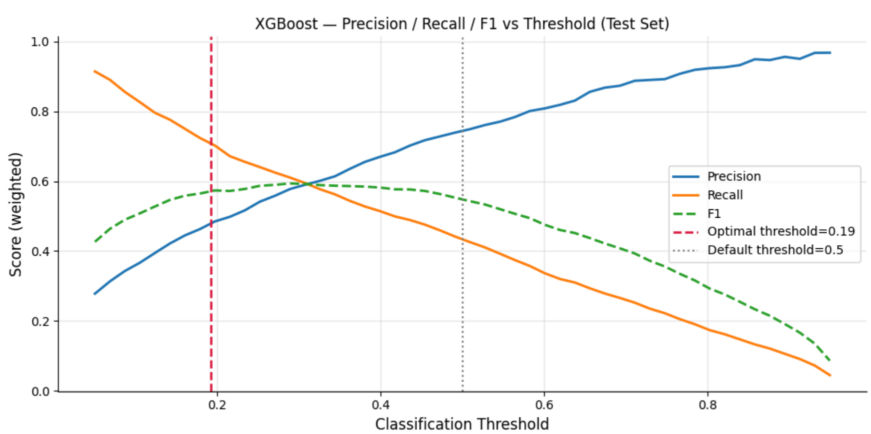

Figure 4. Precision, Recall, and F1 across classification thresholds for XGBoost. At threshold=0.5, recall is very low, the model misses most >50K earners. The optimal threshold of 0.19 (dashed red line) balances precision and recall in a way that maximises profit on the validation set.

b. Profit-Optimised Threshold

Rather than selecting the threshold by maximising F1, which treats false positives and false negatives as equally costly, I used a profit simulation framework that reflects the actual economics of a marketing campaign. The simulation assumes a revenue of $2.50 per successful conversion and a contact cost of $1.00 per person reached. The optimal threshold is the one that maximises expected profit on the validation set, accounting for the population weights.

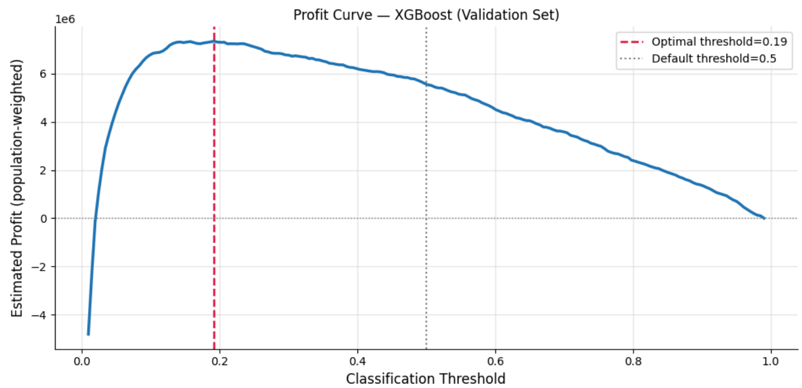

Figure 5. Estimated profit across thresholds on the validation set. The curve peaks at threshold=0.19, after which profit declines as the model contacts too many low-income individuals.

c. Sensitivity Analysis

One legitimate concern about profit-optimised thresholds is that they depend on assumptions about conversion rate that the client may not know precisely. To stress-test the recommendation, I ran the threshold selection across conversion rate assumptions ranging from 5% to 20%.

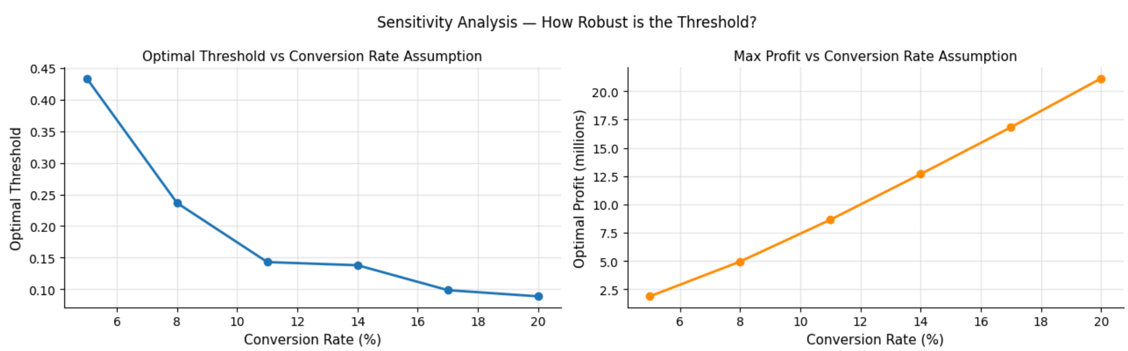

Figure 6. Optimal threshold (left) and maximum profit (right) across different assumed conversion rates.

6. Model Explainability

XGBoost is not a glass-box model, you cannot read off a simple equation that explains any given prediction. However, with SHAP (SHapley Additive exPlanations), we can open the black box and understand exactly which features are driving predictions and in which direction. This matters both for client trust and for knowing what to check if model performance degrades in production.

a. Feature Importance

The dominant feature is major_industry_code_Not_in_universe_or_children, which captures people who are outside the workforce entirely, children, retirees, the unemployed. The model uses this as a first-pass filter to rule out non-earners before considering any other signal. This is operationally important: if production data does not populate this field consistently, model performance will degrade significantly and this is the first thing to check.

Beyond this, the top features are economically intuitive: tax filing status, weeks worked in the year, family structure, occupation type, education level, and capital activity. These are the variables doing the heavy lifting in distinguishing high from low earners.

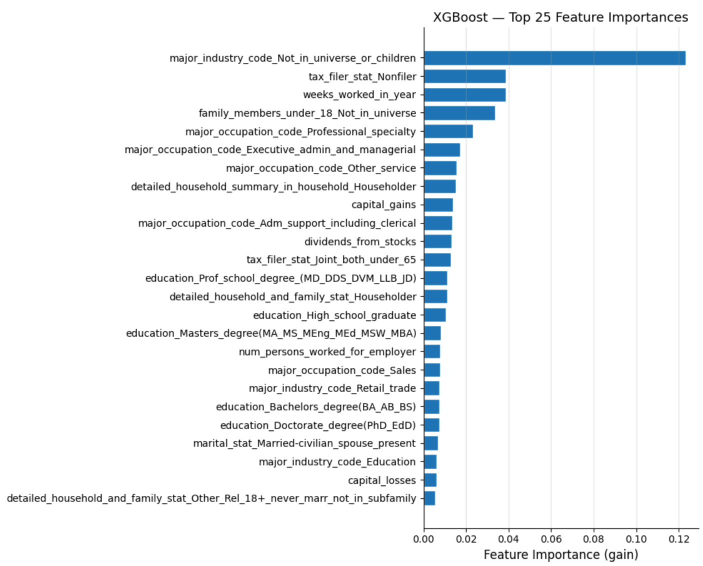

Figure 7. XGBoost feature importances by gain (top 25 features). Gain measures how much each feature improves the model's predictions when it is used in a split.

b. SHAP Analysis

The SHAP analysis tells a cleaner story than raw feature importance. Age and weeks worked in the year are the two most impactful features by a significant margin. Dividends from stocks and capital gains also feature prominently, consistent with our capital_activity_flag finding in the feature engineering section. Education level and occupation type round out the top drivers.

Reading the direction of effects from the beeswarm plot: higher age, more weeks worked, any dividend or capital gain income, and professional or executive occupation codes all push predictions toward the >50K class. Being a non-filer, working in service or clerical roles, and lower educational attainment push in the opposite direction. These patterns are exactly what domain knowledge would predict, which gives confidence that the model is learning genuine structure rather than noise.

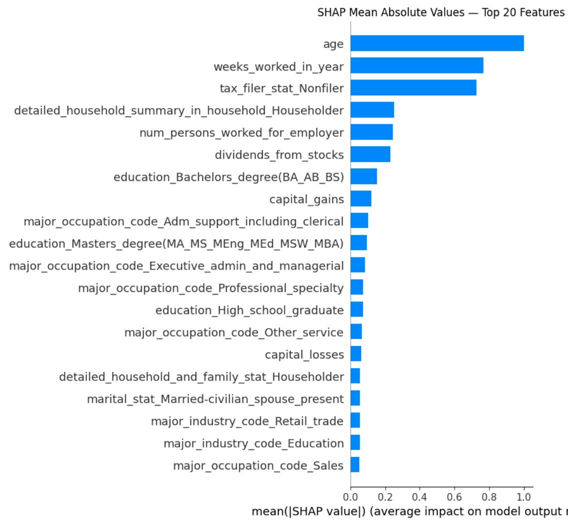

Figure 8. Mean absolute SHAP values for the top 20 features, the average impact each feature has on model predictions across 1,000 test records.

7. Customer Segmentation

Beyond predicting who earns what, a segmentation model was built to group individuals into distinct clusters for targeted marketing. Classification tells you who to reach — segmentation tells you how to talk to them. A single message sent to everyone the classifier flags as high-income is likely to underperform compared to differentiated messaging tailored to meaningfully different subgroups.

K-Means clustering was applied to the preprocessed feature set, producing two models: a global segmentation across the full population using 6 clusters, and a premium segmentation focused specifically on individuals the classifier predicts as high-income using 3 clusters.

a. Global Segmentation

The global segmentation reveals a clear income gradient across clusters. Segment 5 is the most commercially valuable group, with a 19.9% confirmed income rate and 18.1% of the total population. Segment 0 is also worth targeting at 9.4% income rate and 24.9% population share. At the other end, Segments 1, 3, and 4 return income rates of 0.0%, 1.3%, and 0.7% respectively and represent low-value targets for any premium campaign. Segment 1, the largest cluster at 29.1% of the population, is predominantly non-working individuals such as children and retirees, and should be excluded from outreach entirely.

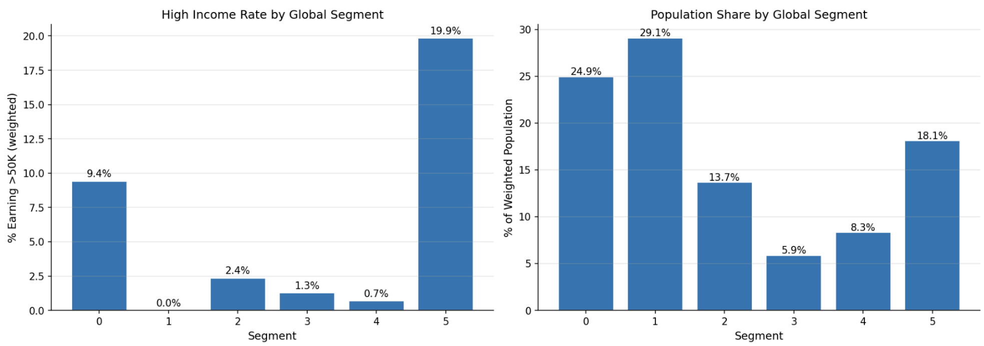

Figure 9. High income rate and population share by global segment.

b. Premium Segmentation

Within the predicted high-income population, all three premium segments show confirmed >50K rates between 50% and 56%, a strong result given the 6.4% base rate in the full population. Segment 1 has the highest precision at 55.7% but accounts for only 4% of the premium group. Segments 0 and 2 together cover 96% of the premium population with confirmation rates of 50.6% and 53.4% respectively.

The clusters differ most clearly on age, wage per hour, capital activity, and weeks worked. Segment 1 shows higher wages and dividend income, pointing to an older investment-active professional profile suited to financial or luxury product messaging. Segment 2 shows strong weeks-worked figures consistent with high-intensity full-time employment, suited to career and productivity oriented campaigns. Segment 0 is the broadest group and the appropriate target for general high-income campaigns.

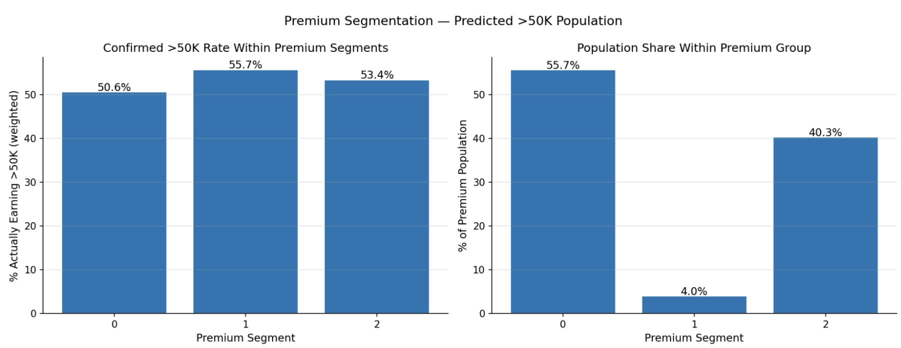

Figure 10. All three segments show strong precision between 50–56%. Segment 0 dominates volume at 55.7% of the premium population.

8. Business Recommendations

Based on the analysis above, here is what I would recommend putting into practice:

• Deploy XGBoost at threshold 0.177. This is the profit-maximising operating point on the validation set. At this threshold, the model targets approximately 9.3% of the population, a manageable contact volume that captures a meaningful proportion of genuine high-income individuals.

• Use the classifier output as a score, not just a binary label. The raw probability score from XGBoost can be used to rank contacts by confidence and prioritise those with scores well above the threshold for premium campaigns.

• Layer segmentation on top of classification. After flagging predicted >50K individuals, use the premium segmentation clusters to differentiate messaging. A single 'high income' label covers a wide range of life stages and financial profiles — segment-specific messaging will outperform a blanket approach.

• Recalibrate the threshold after the first campaign. The current threshold is based on an assumed conversion rate. Once real campaign data is available, re-run the profit simulation with actual figures to update the threshold.

• Conduct a fairness audit before full rollout. Excluding race and sex from the model is a necessary first step but not a complete fairness guarantee. The model may still exhibit disparate impact through correlated features. A proper audit comparing predicted scores and campaign outcomes across demographic groups is strongly recommended before scaling.

• Retrain periodically. This data is from 1994–95. Income distribution patterns shift over time. The model should be retrained whenever substantially newer demographic data becomes available.

References

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